

HDP separation - Proof of concept

Testing Document

Version 1

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General description and purpose

Purpose of test

The findings of test 1 “[Sink-float separation - Proof of concept](#)” have demonstrated the potential of high purities (> 99.9 %) of the two resulting fractions: low-density polymers (LDP) – PP, LDPE, HDPE – and high-density polymers (HDP) – PS, PET, PVC. In the subsequent steps, these fractions must be divided further into its components. This test investigates the next step for the separation of the HDP types, conducted in a similar proof-of-concept fashion (low fidelity, small scale).

Due to the relatively large difference in the density of PS ($\sim 1.050 \text{ g/cm}^3$) compared to PET and PVC ($\sim 1.350 \text{ g/cm}^3$, $\sim 1.400 \text{ g/cm}^3$), the extraction of PS from the HDP mixture offers a good starting point (Bearth, 2025). Hereby, three possible methods are considered:

- **Swirling:**
Adding the HDP mix to water, it will sink to the bottom. When the mixture is then stirred at a very low intensity, the lighter PS is expected to rise while the denser PET and PVC parts stay on the bottom of the container. This allows to skim the PS flakes from the container.
- **Pouring:**
During test 1, it was observed that the PS flakes exited the container during the pouring process first. If the gap between PS and the other materials is large, this could be utilized for separation.
- **Dense sink-float separation:**
The density of the separation medium can be increased by adding salt (NaCl) to the water. According to literature, saline water saturated with salt exhibits a density of roughly 1.2 g/cm^3 at room temperature (David R. Lide, ed., 2005). As the PS flakes are less dense than the separation medium, they should rise to the surface of the container, while the PET and PVC flakes stay on the bottom.

Description of Test

For this test, the color coded plastic flakes from test 1 are used. One mixed test plastic batch is created from the PS, PET, and PVC flakes. The same masses as in test 1 are used of each material (PS: 2.4 g, PET: 2.7 g, PVC: 1.7 g).

As seen in test 1, no substantial change in the separation behavior is exhibited when rerunning the test on the same plastic. Therefore, the batch is first used to test the swirling and pouring methods. Afterwards, it is used for the dense sink-float separation to avoid the contamination of the other plastic with the salt.

The mixed plastic test batch is added to a transparent container with tap water and stirred by a magnetic mixer so that all plastic flakes collect at the bottom. Then, the RPM of the magnetic mixer is adjusted to reach a point at which the PS flakes are still risen while PET and PVC flakes sink to the bottom. The PS flakes are then skimmed with a strainer and dried. Afterwards, the PET and PVC flakes are removed from the container and dried as well. By visually sorting the correctly and incorrectly separated flakes and measuring the resulting fractions' masses, the purity of the separation process can be calculated.

For the pouring test, the plastic test batch is added to the water again and wetted so that all flakes sink to the bottom. Then, the flake behavior is inspected during the pouring process into a strainer to see if the observation from test 1 is reproducible. If so, the PS flakes are decanted and placed on a paper towel to dry. Sorting the plastic types and recording the mass allows the calculation of the purities of the fractions.

Lastly, a separation solution is created by adding salt to tap water until saturated. The density of the solution should be around 1.2 g/cm^3 . A clear container is filled with the solution before the mixed plastic test batch is added and stirred again. After stirring for 3 minutes at 300 RPM, the magnetic mixer is turned off, to allow the material to settle for 2 minutes. The floating fraction is skimmed with a strainer and rinsed under fresh water to remove the salt water from the plastic. The same is done for the sink fraction and both are placed on paper towels to dry. Then, the fractions are sorted into the plastic types to measure the masses and calculate the purities.

Calculations

NaCl water saturation (at 26 wt%):

$$\frac{m(\text{NaCl})}{m(\text{H}_2\text{O}) + m(\text{NaCl})} = 26\% \rightarrow m(\text{NaCl}) = \frac{0.26 m(\text{H}_2\text{O})}{0.74} \approx 0.35 m(\text{H}_2\text{O})$$

$$\Rightarrow m_{\max}(\text{NaCl}(aq)) = 0.35 \frac{kg_{\text{NaCl}}}{l_{\text{H}_2\text{O}}}$$

Plastic separation results:

$$\text{Purity } \varphi_{\text{float}}: \quad \varphi_{\text{float}} = m_{\text{floatPS}} / m_{\text{float}}$$

$$\text{Purity } \varphi_{\text{sink}}: \quad \varphi_{\text{sink}} = m_{\text{sinkPET+PVC}} / m_{\text{sink}}$$

$$\text{Purity } \varphi_{\text{combined}}: \quad \varphi_{\text{combined}} = (\varphi_{\text{sink}} + \varphi_{\text{float}}) / 2$$

Schedule of test with all steps required

1. Prepare batch of mixed test plastics (**PS: 2.4 g, PET: 2.7 g, PVC: 1.7 g**).
2. Prepare transparent container as testing environment for test **a** with **300 ml** of tap water.
3. Add a magnetic stir bar and place the container on a magnetic mixer.
4. Set up camera to record the separation behavior over the test timeframe and start recording.
5. Start the mixer at **500 rpm**. Add the mixed test plastic batch to the test container.
6. Occasionally, turn off the mixer to check if all plastic flakes sink to the bottom.
7. Once all flakes sink, slowly turn on the mixer and incrementally increase the RPM to find a point at which the PS flakes start rising in the current.
8. Leave the mixer at that RPM setting for **30 seconds**.
9. Use the strainer to collect the floating flakes. Place the flakes on a paper towel to dry.
10. Pour the remaining content of the container through the strainer to extract the sink fraction and place on a paper towel as well.
11. After drying off the flakes, sort the float and sink fractions by color and measure the masses.
12. Combine the fractions into a mixed test plastic batch again.

13. Repeat steps 2-6. (Test b)
14. Once all flakes sink, remove the magnetic stir bar.
15. Pour the content of the container into the strainer. Stop after the majority of the PS flakes have entered the strainer. Place the flakes on a paper towel to dry.
16. Pour the remaining content of the container through the strainer to extract the sink fraction and place on a paper towel as well.
17. After drying off the flakes, sort the float and sink fractions by color and measure the masses.
18. Combine the fractions into a mixed test plastic batch again.
19. In the large container, prepare **400 ml** of tap water.
20. Add **140 g** of table salt to the water and stir until fully dissolved.
21. Add additional **10 g** of table salt to the solution and stir. Even after long stirring / resting, the salt should not fully dissolve.
22. Carefully pour **300 ml** of the solution into the transparent container so that no undissolved salt is transferred. (Test c)
23. Add a magnetic stir bar and place the container on a magnetic mixer.
24. Set up camera to record the separation behavior over the test timeframe and start recording.
25. Start the mixer at **300 rpm** for 3 minutes. Add mixed test plastic batch to the test container.
26. After 3 minutes, turn of the magnetic mixer and let the plastic flakes settle for 2 minutes.
27. Use the strainer to collect the float fraction, carefully rinse with tap water to remove the salt, and place on a paper towel to dry.
28. Pour the content of the container trough the strainer back into the large container to collect the saline water and to remove the sink fraction.
29. Carefully rinse the sink fraction with tap water to remove the salt, and place on a paper towel to dry.
30. After drying off the flakes, sort the float and sink fractions by color and measure the masses.

Risk analysis

Potential risks associated with this test

None.

Mitigation steps

Safety glasses are worn during the whole process.

Risks after taking mitigation steps

None.

Tools and devices required

Tool	Check-in tick
1x safety glasses	X
Paper towels	X

1x laptop	X
1x camera (phone)	X
1x timer (laptop or phone can be used)	X
1x scale	X
Containers of cut plastic flakes (PS, PET, PVC)	X
1x container for mixed plastic test batches	X
1x transparent container for tests	X
1x large container (capacity ≥ 0.5 l)	X
1x magnetic stir bar	X
1x magnetic mixer	X
1x strainer	X
1 l tap water	X
150 g NaCl	X
1x tweezers	X
1x calculator	X

Testing Results

All recordings of the separation process are available [here](#).

Swirling

a: 10 s, most flakes submerged; 30 s no change; 1 min no change; turbulent stirring & manual submerging needed for complete submersion

Test	Input masses [g]			Float-output masses [g]			Sink-output masses [g]		
a	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
(160 rpm)	2.4	2.7	1.7	0.76	0.03	0	1.63	4.23	
	Tot: 6.8 (theoretically 6.8)								

Any unexpected behaviors:

- PS flakes tend to float higher, but mix with PET on lower levels
- most material stays at low level (highest velocity due to magnetic stir bar at the bottom?)
- very difficult to extract
- impulses improve separation, PET sinks faster after impulse

Pouring

Test	Input masses [g]			Float-output masses [g]			Sink-output masses [g]		
b	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	0.73	0.03	0	1.65	4.27	
	Tot: 6.8 (theoretically 6.8)								

Any unexpected behaviors:

- difference in exact total output weights between tests a and b could be remaining water on the flakes

Dense sink-float separation

Separation solution:

$m(\text{H}_2\text{O}) = 400 \text{ g}$,

$m(\text{NaCl}) = 140 \text{ g}$ (enough for saturation, very little residual) + 10 g (much residual)

Test	Input masses [g]			Float-output masses [g]			Sink-output masses [g]		
c	PS	PET	PVC	PS	PET	PVC	PS	PET	PVC
	2.4	2.7	1.7	2.39	0.08	0	0	4.20	
	Tot: 6.8 (theoretically 6.8)								

Any unexpected behaviors:

- the RPM was increased to 500, since the higher density of the medium required it
- one PET flake remaining in float has a pocket with air in it (good shredding is crucial)
- after drying, during weighting, the PET flakes were electrostatically charged

Evaluation

The swirling and pouring methods did not display sufficiently high purities compared to the dense sink-float separation, as seen in Table 1.

One potential source of failure is the magnetic stir bar being located at the bottom of the container. This results in the highest velocity of the separation medium to occur at the bottom as well, where most of the flakes are then collected, making an extraction of just one material impossible. Moving the stirrer to the top or using pulsed actuation could improve the separation since PET and PVC flakes sink much faster in water than PS.

The pouring method exhibits very limited control of the separation with most flakes staying at the bottom of the container. This results in a good purity of the float fraction but poor values for the sink fraction, an especially unfavorable outcome since the sink fraction requires a subsequent sorting step additionally affecting the overall purity.

The best results are obtained with the dense sink-float separation. The clear separation by flake density is promoted by a lack of dynamic behaviors, allowing the flakes to separate with time. However, good shredding as a preparatory step is crucial as one PET flake remained in the float fraction due to an air bubble being caught in a small pocket of the flake skewing the density of the particle. Furthermore, this method has the disadvantage of making a high amount of salt necessary ($0.35 \frac{\text{kg}_{\text{NaCl}}}{\text{l}_{\text{H}_2\text{O}}}$). These costs of roughly MWK 1000 per kilogram must be accounted for in Blantyre (“Malawi Table Salt Prices,” n.d.).

Table 1: Purity results sorted by combined purity.

Test	Purity float	Purity sink	Purity combined
c	96.76%	100.00%	98.38%
a	96.20%	72.18%	84.19%
b	96.05%	72.13%	84.09%

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